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Organisms and Populations

NEET Biology · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	124 questions
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MCQ (Single Correct Answer) 124 questions

Q1. RE-NEET 2026

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason **R**.

Assertion A : Abingdon tortoise in Galapagos islands became extinct within a decade after goats were introduced.

Reason R: Goats were more efficient at browsing than Abingdon tortoise.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) **A** is not correct but **R** is correct

(B) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(C) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

(D) **A** is correct but **R** is not correct

Q2. RE-NEET 2026

Open solution

Consider a population of 10 million cells. Given the per-capita birth rate of 0.002 (per unit time) and the percapita death rate of 0.002 (per unit time), the expected number of cells after 10 generations is ____ .

(A) 100 million (B) 1 million (C) 5 million (D) 10 million

Q3. RE-NEET 2026

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : The logistic growth model of populations is considered more realistic than the exponential growth model.

Reason R: Resources are finite.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) **A** is not correct but **R** is correct

(B) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(C) Both **A** and **R** are correct but **R** is not the correct explanation of **A**

(D) **A** is correct but **R** is not correct

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Q4. RE-NEET 2026

Open solution

Match List-I with List-II.

	List-I		List-II
A.	Both species are harmed	I.	Predation
B.	One species is harmed and the other is benefited	II.	Mutualism
C.	Both species are benefited	III.	Competition
D.	One is benefited while the other has no effect	IV.	Commensalism

Choose the correct answer from the options given below:

(A) A-III, B-I, C-II, D-IV
(B) A-III, B-IV, C-II, D-I
(C) A-I, B-II, C-III, D-IV
(D) A-II, B-I, C-IV, D-III

Q5. NEET 2026

Open solution

Which of the following equations depicts Verhulst-Pearl logistic population growth?

(A) $\frac{dN}{dt} = rN\left(\frac{K+N}{K}\right)$.
(B) $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$
(C) $\frac{dN}{dt} = rN\left(\frac{K-N}{N}\right)$
(D) $\frac{dN}{dt} = rN\left(\frac{K}{K-N}\right)$

Q6. NEET 2026

Open solution

Which one of the following is an appropriate example of sexual deceit?

(A) Female wasp and fig
(B) Ophrys and bumblebee
(C) Sea anemone and clown fish
(D) Cuckoo and crow

Q7. NEET 2026

Open solution

Choose the correct statements regarding population interactions between two species.

A. In both parasitism and commensalism, only one species benefits and the other species is harmed.

B. Both species benefit in mutualism.

C. Both species benefit in commensalism.

D. In parasitism, only one species benefits and the other species is harmed.

E. In amensalism, one species is harmed and the other is unaffected.

Choose the correct answer from the options given below:

(A) B and E only
(B) A and B only
(C) B, D and E only
(D) A and D only

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Q8. NEET 2025

Epiphytes that are growing on a mango branch is an example of which of the following?

(A) Predation (B) Amensalism (C) Commensalism (D) Mutualism

Open solution

Q9. NEET 2025

Given below are two statements :

Statement I : Fig fruit is a non-vegetarian fruit as it has enclosed fig wasps in it.

Statement II : Fig wasp and fig tree exhibit mutual relationship as fig wasp completes its life cycle in fig fruit and fig fruit gets pollinated by fig wasp.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Statement I is correct but statement II is incorrect

(B) Statement I is incorrect but statement II is correct

(C) Both statement I and statement II are correct

(D) Both statement I and statement II are incorrect

Open solution

Q10. NEET 2025

Which one of the following equations represents the Verhulst-Pearl Logistic Growth of population?

(A) $\frac{dN}{dt} = rN\left(\frac{N-K}{N}\right)$ (B) $\frac{dN}{dt} = N\left(\frac{r-K}{K}\right)$

(C) $\frac{dN}{dt} = r\left(\frac{K-N}{K}\right)$ (D) $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$

Open solution

Q11. NEET 2024 (Re-Examination)

When will the population density increase, under special conditions? When the number of :

(A) Deaths exceeds number of births and also number of emigrants equals number of immigrants.

(B) Births plus number of immigrants equals number of deaths plus number of emigrants.

(C) Births plus number of emigrants is more than the number of deaths plus number of immigrants.

(D) Births plus number of immigrants is more than the sum of number of deaths and number of emigrants.

Open solution

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Q12. NEET 2024 (Re-Examination)

Open solution

Match List-I with List-II:

	List - I		List - II
(A)	Predator	(I)	Ophrys
(B)	Mutualism	(II)	Pisaster
(C)	Parasitism	(III)	Female wasp and fig
(D)	Sexual deceit	(IV)	Plasmodium

Choose the correct answer from the options given below:

(A) A-III, B-II, C-I, D-IV

(B) A-IV, B-I, C-II, D-III

(C) A-II, B-III, C-I, D-IV

(D) A-II, B-III, C-IV, D-I

Q13. NEET 2024 (Re-Examination)

Open solution

Match List-I with List-II.

	List - I		List - II
(A)	Migratory flamingoes and resident fish in South American lakes	(I)	Interference competition
(B)	Abingdon tortoise became extinct after introduction of goats in their habitat	(II)	Competitive release
(C)	Chthamalus expands its distributional range in the absence of Balanus	(III)	Resource Partitioning
(D)	Five closely related species of Warblers feeding in different locations on same tree	(IV)	Interspecific competition

Choose the correct answer from the options given below:

(A) A-I, B-IV, C-III, D-II

(B) A-IV, B-I, C-II, D-III

(C) A-III, B-I, C-II, D-IV

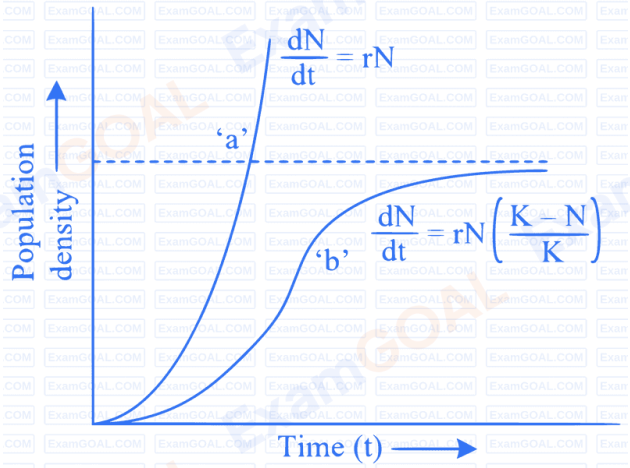
(D) A-II, B-IV, C-III, D-I

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Q14. NEET 2024 (Re-Examination)

Open solution

What do 'a' and 'b' represent in the following population growth curve?



(A) 'a' represents exponential growth when resources are not limiting the growth; and 'b' represents logistic growth when resources are limiting the growth.

(B) 'a' represents logistic growth when resources are not limiting the growth; 'b' represents exponential growth when resources are limiting the growth.

(C) 'a' represents carrying capacity and 'b' shows logistic growth when resources are limiting the growth.

(D) 'a' represents exponential growth when resources are not limiting the growth and 'b' shows carrying capacity.

Q15. NEET 2024

Open solution

The equation of Verhulst-Pearl logistic growth is $\frac{dN}{dt} = rN\left[\frac{K-N}{K}\right]$. From this equation, K indicates:

(A) Intrinsic rate of natural increase

(B) Biotic potential

(C) Carrying capacity

(D) Population density

Q16. NEET 2024

Open solution

Given below are two statements:

Statement I: Gause's competitive exclusion principle states that two closely related species competing for different resources cannot exist indefinitely.

Statement II: According to Gause's principle, during competition, the inferior will be eliminated. This may be true if resources are limiting.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

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Q17. NEET 2023 Manipur

Plants offer rewards to animals in the form of pollen and nectar and the animals facilitate the pollination process. This is an example of :

(A) Amensalism (B) Competition (C) Commensalism (D) Mutualism

Open solution

Q18. NEET 2023 Manipur

For chemical defence against herbivores, Calotropis has :

(A) cardiac glycosides (B) strychnine

(C) toxic ricin (D) distasteful quinine

Open solution

Q19. NEET 2023 Manipur

If there are 250 snails in a pond, and within a year their number increases to 2500 by reproduction. What should be their birth rate per snail per year ?

(A) 10 (B) 9 (C) 25 (D) 15

Open solution

Q20. NEET 2023

Given below are two statements:

Statement I : Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and competitively inferior one will be eliminated eventually.

Statement II : In general, carnivores are more adversely affected by competition than herbivores.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are false.

(B) Statement I is correct but Statement II is false.

(C) Statement I is incorrect but Statement II is true.

(D) Both Statement I and Statement II are true.

Open solution

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Q21. NEET 2023

Open solution

Match List I with List II:

	List I (Interaction)		List II (Species A and B)
(A)	Mutualism	(I)	+(A), 0(B)
(B)	Commensalism	(II)	−(A), 0(B)
(C)	Amensalism	(III)	+(A), −(B)
(D)	Parasitism	(IV)	+(A), +(B)

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-II, D-III

(B) A-IV, B-III, C-I, D-II

(C) A-III, B-I, C-IV, D-II

(D) A-IV, B-II, C-I, D-III

Q22. NEET 2023

Open solution

Match List I with List II.

	List I (Interacting species)		List II (Name of interaction)
(A)	A Leopard and a Lion in a forest/grassland	(I)	Competition
(B)	A Cuckoo laying egg in a Crow's nest	(II)	Brood parasitism
(C)	Fungi and root of a higher plant in Mycorrhizae	(III)	Mutualism
(D)	A cattle egret and a Cattle in a field	(IV)	Commensalism

Choose the correct answer from the options given below.

(A) A-I, B-II, C-IV, D-III

(B) A-III, B-IV, C-I, D-II

(C) A-II, B-III, C-I, D-IV

(D) A-I, B-II, C-III, D-IV

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Q23. NEET 2023

Open solution

Match List I with List II.

	List I		List II
(A)	Logistic growth	(I)	Unlimited resource availability condition
(B)	Exponential growth	(II)	Limited resource availability condition
(C)	Expanding age pyramid	(III)	The percent individuals of pre-reproductive age is largest followed by reproductive and post reproductive age groups
(D)	Stable age pyramid	(IV)	The percent individuals of pre-reproductive and reproductive age group are same

Choose the correct answer from the options given below:

(A) A-II, B-III, C-I, D-IV
(B) A-II, B-IV, C-I, D-III
(C) A-II, B-IV, C-III, D-I
(D) A-II, B-I, C-III, D-IV

Q24. NEET 2022 Phase 2

Open solution

Two butterfly species are competing for the same nectar of a flower in a garden. To survive and coexist together, they may avoid competition in the same garden by :

(A) Predating on each other
(B) Feeding at the same time
(C) Choosing different foraging patterns
(D) Increasing time spent on attacking each other

Q25. NEET 2022 Phase 1

Open solution

Which one of the following statements cannot be connected to Predation?

(A) It helps in maintaining species diversity in a community
(B) It might lead to extinction of a species
(C) Both the integrating species are negatively impacted
(D) It is necessitated by nature to maintain the ecological balance

Q26. NEET 2022 Phase 1

Open solution

While explaining interspecific interaction of population, (+) sign is assigned for beneficial interaction, (–) sign is assigned for detrimental interaction and (0) for neutral interaction. Which of the following interactions can be assigned (+) for one species and (–) for another species involved in the interaction?

(A) Predation **(B)** Amensalism **(C)** Commensalism **(D)** Competition

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Q27. NEET 2022 Phase 1

If '8' Drosophila in a laboratory population of '80' died during a week, the death rate in the population is _____ individuals per Drosophila per week.

(A) 0.1 (B) 10 (C) 1.0 (D) zero

Open solution

Q28. NEET 2021

Inspite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival?

(A) Predation
(B) Resource partitioning
(C) Competitive release
(D) Mutualism

Open solution

Q29. NEET 2021

Amensalism can be represented as :

(A) Species A (+) ; Species B (0)
(B) Species A (–); Species B (0)
(C) Species A (+); Species B (+)
(D) Species A (–); Species B (–)

Open solution

Q30. NEET 2021

In the exponential growth equation
$N_t = N_0 e^{rt}$, e represents

(A) The base of geometric logarithms
(B) The base of number logarithms
(C) The base of exponential logarithms
(D) The base of natural logarithms

Open solution

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Q31. NEET 2021

Match List-I with List-II

Open solution

	List-I		List-II
(a)	Allen's Rule	(i)	Kangaroo rat
(b)	Physiological adaptation	(ii)	Desert lizard
(c)	Behavioural adaptation	(iii)	Marine fish at depth
(d)	Biochemical adaptation	(iv)	Polar seal

Choose the correct answer from the options given below.

(A)

(a)	(b)	(c)	(d)
(iv)	(iii)	(ii)	(i)

(B)

(a)	(b)	(c)	(d)
(iv)	(ii)	(iii)	(i)

(C)

(a)	(b)	(c)	(d)
(iv)	(i)	(iii)	(ii)

(D)

(a)	(b)	(c)	(d)
(iv)	(i)	(ii)	(iii)

Q32. NEET 2020 Phase 2

The impact of immigration on population density is :

Open solution

(A) Positive
(B) Negative
(C) Both positive and negative
(D) Neutralized by natality

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Q33. NEET 2020 Phase 2

Match the items in Column I with those in Column II :

Column I	Column II
(a) Herbivores-Plants	(i) Commensalism
(b) Mycorrhiza-Plants	(ii) Mutualism
(c) Sheep-Cattle	(iii) Predation
(d) Orchid-Tree	(iv) Competition

Select the correct option from following :

(A) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
(B) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
(C) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
(D) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

Open solution

Q34. NEET 2020 Phase 1

Secondary metabolites such as nicotine, strychnine and caffeine are produced by plants for their

(A) Nutritive value
(B) Growth response
(C) Effect on reproduction
(D) Defence action

Open solution

Q35. NEET 2020 Phase 1

Which of the following is not an attribute of a population?

(A) Natality
(B) Sex ratio
(C) Mortality
(D) Species interaction

Open solution

Q36. NEET 2019 (Odisha)

Between which among the following, the relationship is **not** an example of commensalism?

(A) Female wasp and fig species
(B) Orchid and the tree on which it grows
(C) Cattle Egret and grazing cattle
(D) Sea Anemone and Clown fish

Open solution

Q37. NEET 2019 (Odisha)

Carnivorous animals - lions and leopards, occupy the same niche but lions predate mostly larger animals and leopards take smaller ones. This mechanism of competition is referred to as–

(A) Competitive exclusion
(B) Character displacement
(C) Altruism
(D) Resource partitioning

Open solution

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Q38. NEET 2019

Open solution

Match Column-I with Column –II.

Column - I	Column - II
(a) Saprophyte	(i) Symbiotic association of fungi with plant roots
(b) Parasite	(ii) Decomposition of dead organic materials
(c) Lichens	(iii) Living on living plants or animals
(d) Mycorrhiza	(iv) Symbiotic association of algae and fungi

Choose the **correct** answer from the options given below :

(A)

(a)	(b)	(c)	(d)
(i)	(ii)	(iii)	(iv)

(B)

(a)	(b)	(c)	(d)
(iii)	(ii)	(i)	(iv)

(C)

(a)	(b)	(c)	(d)
(ii)	(i)	(iii)	(iv)

(D)

(a)	(b)	(c)	(d)
(ii)	(iii)	(iv)	(i)

Q39. NEET 2018

Open solution

Niche is

(A) all the biological factors in the organism's environment

(B) the physical space where an organism lives

(C) the range of temperature that the organism needs to live

(D) the functional role played by the organism where it lives

Q40. NEET 2018

Open solution

Which one of the following population interactions is widely used in medical science for the production of antibiotics?

(A) Commensalism (B) Mutualism (C) Parasitism (D) Amensalism

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Q41. NEET 2018

Natality refers to

(A) death rate

(B) birth rate

(C) number of individuals leaving the habitat

(D) number of individuals entering a habitat

Open solution

Q42. NEET 2018

Pneumatophores occur in

(A) halophytes

(B) free-floating hydrophytes

(C) carnivorous plants

(D) submerged hydrophytes

Open solution

Q43. NEET 2018

Which one of the following plants shows a very close relationship with a species of moth, where none of the two can complete its life cycle without the other?

(A) Hydrilla (B) Yucca (C) Banana (D) Viola

Open solution

Q44. NEET 2018

In a growing population of a country,

(A) pre-reproductive individuals are more than the reproductive individuals

(B) reproductive individuals are less than the post-reproductive individuals

(C) reproductive and pre-reproductive individuals are equal in number

(D) pre-reproductive individuals are less than the reproductive individuals.

Open solution

Q45. NEET 2017

Asymptote in a logistic growth curve is obtained when :

(A) K > N

(B) The value of 'r' approaches zero

(C) K< N

(D) K = N

Open solution

Q46. NEET 2017

Plants which produce characteristic pneumatophores and show vivipary belong to :

(A) Halophytes (B) Psammophytes (C) Hydrophytes (D) Mesophytes

Open solution

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Q47. NEET 2017

Mycorrhizae are the example of :

(A) Fungistasis (B) Antibiosis (C) Mutualism (D) Amensalism

Open solution

Q48. NEET 2016 Phase 2

Which of the following is correct for r-selected species ?

(A) Small number of progeny with small size
(B) Small number of progeny with large size
(C) Large number of progeny with small size
(D) Large number of progeny with large size

Open solution

Q49. NEET 2016 Phase 2

If '+' sign is assigned to beneficial interaction, '–' sign to detrimental and '0' sign to neutral interaction, then the population interaction represented by '+' '–' refers to

(A) parasitism (B) commensalism (C) mutualism (D) amensalism

Open solution

Q50. NEET 2016 Phase 2

The principle of competitive exclusion was stated by

(A) MacArthur (B) G. F. Gause (C) Verhulst & Pearl (D) C. Darwin

Open solution

Q51. NEET 2016 Phase 1

It is much easier for a small animal to run uphill than for a large animal, because -

(A) The efficiency of muscles in large animals is less than in the small animals
(B) Smaller animals have a higher metabolic rate
(C) It is easier to carry a small body weight
(D) Small animals have a lower O_2 requirement

Open solution

Q52. NEET 2016 Phase 1

When does the growth rate of a population following the logistic model equal zero? The logistic model is given as dN/dt = rN(1-N/K)

(A) When death rate is greater than birth rate.
(B) When N/K equals zero.
(C) When N/K is exactly one.
(D) When N nears the carrying capacity of the habitat.

Open solution

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Q53. NEET 2016 Phase 1

Gause's principle of competitive exclusion states that -

(A) Larger organisms exclude smaller ones through competition

(B) More abundant species will exclude the less abundant species through competition

(C) Competition for the same resources excludes species having different food preferences

(D) No two species can occupy the same niche indefinitely for the same limiting resources

Open solution

Q54. AIPMT 2015

In which of the following interactions both partners are adversely affected?

(A) Competition (B) Predation (C) Mutualism (D) Parasitism

Open solution

Q55. AIPMT 2015

An association of individuals of different species living in the same habitat and having functional interactions is -

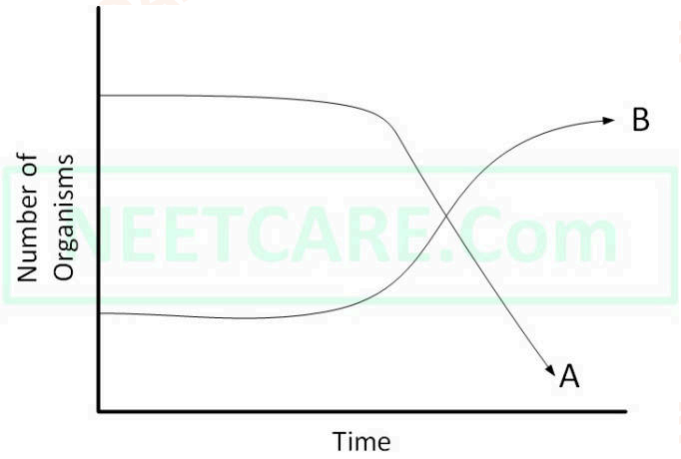
(A) Ecological niche (B) Ecosystem (C) Population (D) Biotic community

Open solution

Q56. AIPMT 2015 Cancelled Paper

The following graph depicts changes in two populations (A and B) of herbivores in a grassy field. A possible reason for these changes is that:

Open solution



(A) Population B competed more successfully for food than population A

(B) Population A consumed the members of population B

(C) Both plant populations in this habitat decreased

(D) Population A produced more offspring than population B

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Q57. AIPMT 2014

Just as a person moving from Delhi to Shimla to escape the heat for the duration of hot summer, thousands of migratory birds from Siberia and other extremely cold northern regions move to:

(A) Keolado National Park
(B) Meghalaya
(C) Corbett National Park
(D) Western Ghat

Open solution

Q58. NEET 2013 (Karnataka)

The age pyramid with broad base indicates :

(A) high percentage of young individuals
(B) high percentage of old individuals
(C) a stable population
(D) low percentage of young individuals

Open solution

Q59. NEET 2013 (Karnataka)

Which one of the following is not a parasitic adaptation ?

(A) Loss of digestive organs
(B) Loss of unnecessary sense organs
(C) Development of adhesive organs
(D) Loss of reproductive capacity

Open solution

Q60. NEET 2013 (Karnataka)

Benthic organisms are affected most by :

(A) Surface turbulence of water
(B) Water-holding capacity of soil
(C) Light reaching the forest floor
(D) Sediment characteristics of aquatic ecosystems

Open solution

Q61. NEET 2013

A sedentary sea anemone gets attached to the shell lining of hermit crab. The association is :

(A) Commensalism (B) Amensalism (C) Ectoparasitism (D) Symbiosis

Open solution

Q62. NEET 2013

A biologist studied the population of rats in a barn, He found that the average natality was 250, average mortality 240, immigration 20 and emigration 30. The net increase in population is :

(A) 15 (B) 05 (C) 10 (D) zero

Open solution

Q63. AIPMT 2012 Mains

Cuscuta is an example of :

(A) Ectoparasitism (B) Predation (C) Endoparasitism (D) Brood parasitism

Open solution

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Q64. AIPMT 2011 Mains

The logistic population growth is expressed by the equation :

(A) $dt/dN = Nr\left(\frac{K-N}{K}\right)$

(B) $dN/dt = rN\left(\frac{K-N}{K}\right)$

(C) $dN/dt = rN\left(\frac{N-K}{N}\right)$

(D) $dN/dt = rN$

Open solution

Q65. AIPMT 2011 Mains

Consider the following statements (A)-(D) each with one or two blanks :

(A) Bears go into _____ (1) ____ during winter to _____(2)____ cold weather.

(B) A conical age pyramid with a broad base represents ____(3)_____ human population.

(C) A wasp pollinating a fig flower is an example of _____ (4) _____ .

(D) An area with high levels of species richness is known as ____(5)_____.

Which of the following options, gives the correct fill ups for the respective blank numbers from (1) to (5) in the statements ?

(A) (1) - aestivation, (2) - escape,
(3) - stable, (4) - mutualism

(B) (3) - expanding, (4) - commensalism,
(5) - biodiversity park

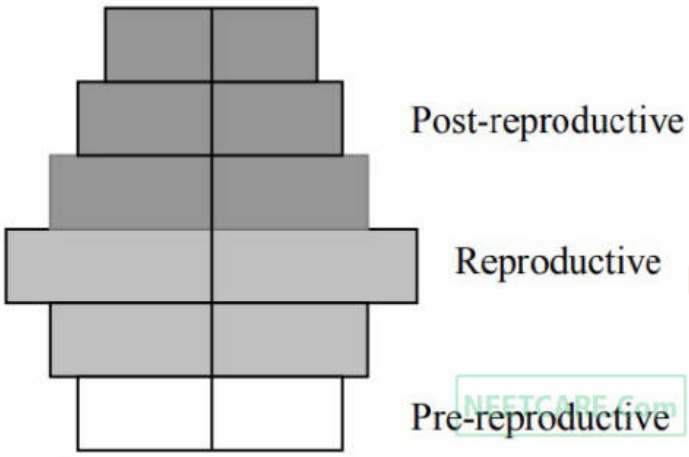
(C) (1) - hibernation, (2) - escape,
(3) - expanding, (5) - hot spot

(D) (3) - stable, (4) - commensalism,
(5) - marsh

Open solution

Q66. AIPMT 2011 Prelims

What type of human population is represented by the following age pyramid ?



(A) Declining population

(B) Vanishing population

(C) Expanding population

(D) Stable population

Open solution

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Q67. AIPMT 2011 Prelims

Large Woody Vines are more commonly found in

(A) Mangroves
(B) Tropical rainforests
(C) Alpine forests
(D) Temperate forests

Open solution

Q68. AIPMT 2011 Prelims

Consider the following four conditions (a – d) and select the correct pair of them as adaptation to environment in desert lizards.
The conditions :
(a) burrowing in soil to escape high temperature
(b) losing heat rapidly from the body during high temperature
(c) bask in sun when temperature is low
(d) insulating body due to thick fatty dermis

(A) (c), (d)
(B) (b), (d)
(C) (a), (c)
(D) (a), (b)

Open solution

Q69. AIPMT 2011 Prelims

Which one of the following is categorised as a parasite in true sense ?

(A) The female anopheles bites and sucks blood from humans
(B) Head louse living on the human scalp as well as laying eggs on human hair
(C) The cuckoo (koel) lays its eggs in crow's nest
(D) Human foetus developing inside the uterus draws nourishment from the mother

Open solution

Q70. AIPMT 2010 Mains

Which one of the following is most appropriately defined ?

(A) Parasite is an organism which always lives inside the body of other organism and may kill it.
(B) Host is an organism which provides food to another organism.
(C) Amensalism is a relationship in which one species is benefited where as the other is unaffected
(D) Predator is an organism that catches and kills other organism for food.

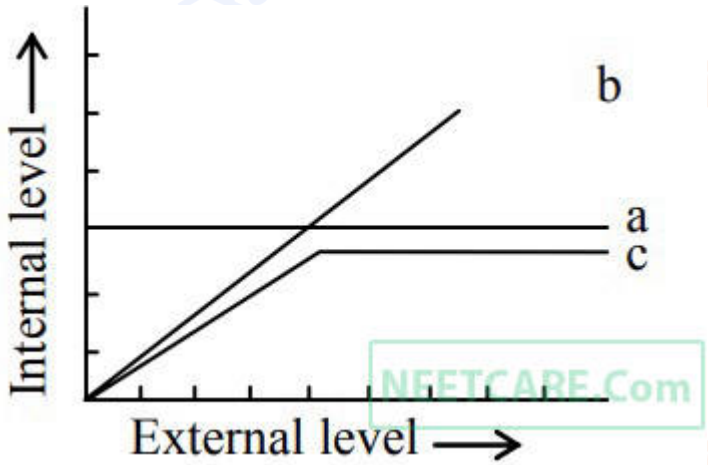
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Q71. AIPMT 2010 Prelims

Open solution

The figure given below is a diagrammatic representation of response of organisms to abiotic factors. What do a, b and c represent respectively –



(A) (a) → partial regulator, (b) → regulator, (c) → conformer

(B) (a) → conformer, (b) → regulator, (c) → partial regulator

(C) (a) → regulator, (b) → conformer, (c) → partial regulator

(D) (a) → regulator, (b) → partial regulator, (c) → conformer

Q72. AIPMT 2010 Prelims

Open solution

Study the four statements (a–d) given below and select the two correct ones out of them –
(a) A lion eating a deer and a sparrow feeding on grain are ecologically similar in being consumers
(b) Predator star fish Pisaster helps in maintaining species diversity of some invertebrates
(c) Predators ultimately lead to the extinction of prey species
(d) Production of chemicals such as nicotine, strychnine by the plants are metabolic disorders
The two correct statements are-

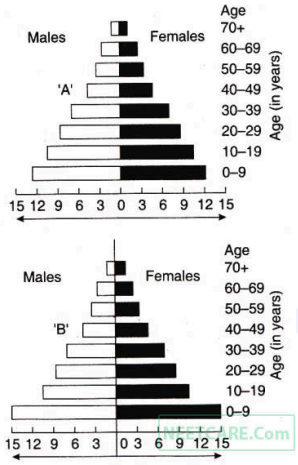
(A) b and c (B) a and d (C) c and d (D) a and b

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Q73. AIPMT 2009

A country with a high rate of population growth took measures to reduce it. The figure below shows age-sex pyramids of populations A and B twenty years apart. Select the correct interpretation about them : –

Open solution



(A) "A" is more recent shows slight reduction in the growth rate

(B) "B" is more recent showing that population is very young

(C) "A" is the earlier pyramid and no change has occurred in the growth rate

(D) "B" is earlier pyramid and shows stabilized growth rate

Q74. AIPMT 2009

Reduction in vascular tissue, mechanical tissue and cuticle is characteristic of :

(A) Xerophytes (B) Epiphytes (C) Hydrophytes (D) Mesophytes

Open solution

Q75. AIPMT 2008

Consider the following four statements (1-4) about certain desert animals such as kangaroo-rat.
(i) They have dark colour and high rate of reproduction and excrete solid urine
(ii) They do not drink water, breathe at a slow rate to conserve water and have their body covered with thick hairs
(iii) They feed on dry seeds and do not required drinking water
(iv) They excrete very concentrated urine and do not use water to regulate body temperature
Out of these four, which two are correct-

(A) (i) and (ii) (B) (ii) and (iii) (C) (iii) and (i) (D) (iii) and (iv)

Open solution

Q76. AIPMT 2008

Quercus species are the dominant component in:

(A) Temperate deciduous forests (B) Alpine forests

(C) Scrub forests (D) Tropical rain forests

Open solution

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Q77. AIPMT 2007

Geometric representation of age structure is a characteristic of :-

(A) Biotic community (B) Landscape (C) Ecosystem (D) Population

Open solution

Q78. AIPMT 2007

The population of an insect species shows an explosive increase in numbers during rainy season followed by its disappearance at the end of the season. What does this show ?

(A) The food plants mature and die at the end of the rainy season

(B) The population of its predators increases enormously

(C) Its population growth curve is of J-type

(D) S-shaped or sigmoid growth of this insect

Open solution

Q79. AIPMT 2007

If the mean and the median pertaining to a certain character of a population are of the same value, the following is most likely to occur:-

(A) A normal distribution (B) a T-shaped curve

(C) A skewed curve (D) A bi-modal distribution

Open solution

Q80. AIPMT 2007

A high density of elephant population in an area can result in :-

(A) Mutualism (B) Interspecific competition

(C) Predation on one another (D) Intraspecific competition

Open solution

Q81. AIPMT 2007

Which of the following is an example of negative feedback loop in humans?

(A) Secretion of tears after falling of sand particles into the eye

(B) Salivation of mouth at the sight of delicious food

(C) Secretion of sweat glands and constriction of skin blood vessels when it is too hot

(D) Constriction of skin blood vessels and contraction of skeletal muscles when it is too cold

Open solution

Q82. AIPMT 2006

The formula for exponential population growth is-

(A) dN/dt = rN (B) dt/dN = rN (C) rN/dN = dt (D) dN/rN = dt

Open solution

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Q83. AIPMT 2006

Niche overlap indicates-

(A) Sharing of one or more resources between the two species

(B) Mutualism between two species

(C) Active cooperation between two species

(D) Two different parasites on the same host

Open solution

Q84. AIPMT 2005

More than 70% of world's fresh water is contained in –

(A) Polar ice

(B) Greenland

(C) Antarctica

(D) Glaciers and Mountains

Open solution

Q85. AIPMT 2005

At which latitude, heat gain through insolation approximately equals heat loss through terrestrial radiation -

(A) 40° North and South

(B) $42\frac{1}{2}^{o}$ North and South

(C) 66° North and South

(D) $22\frac{1}{2}^{o}$ North and South

Open solution

Q86. AIPMT 2005

Animals have the innate ability to escape from predation. Examples for the same are given below. Select the incorrect example

(A) Poison fangs in snakes

(B) Enlargement of body size by swallowing air in puffer fish

(C) Colour change in chameleon

(D) Melanism in moths

Open solution

Q87. AIPMT 2005

Which one of the following pairs in mismatched ?

(A) Tundra - permafrost

(B) Savanna - acacia trees

(C) Prairie - epiphytes

(D) Coniferous forest - evergreen trees

Open solution

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Q88. AIPMT 2004

Certain characteristic demographic features of developing countries are -

(A) High fertility, low or rapidly falling mortality rate, rapid population growth and a very young age distribution

(B) High mortality high density, uneven population growth and a very old age distribution

(C) High fertility, high density, rapidly rising mortality rate and very young age distribution

(D) High infant mortality, low fertility, uneven population growth and a very young age distribution

Open solution

Q89. AIPMT 2004

What is a keystone species ?

(A) A dominant species that constitutes a large proportion of the biomass and which affects many other species

(B) A species which makes up only a small proportion of the total biomass of a community, yet has a huge impact on the community's organization and survival

(C) A common species that has plenty of biomass, yet has a fairly low impact on the community's organization

(D) A rare species that has minimal impact on the biomass and on other species in the community

Open solution

Q90. AIPMT 2004

In which of the following pairs is the specific characteristic of a soil not correctly matched ?

(A) Chernozems – Richest soil in the world

(B) Laterite – Contains aluminium compound

(C) Terra rossa – Most suitable for roses

(D) Black soil – Rich in calcium carbonate

Open solution

Q91. AIPMT 2004

The maximum growth rate occurs in :-

(A) Exponential phase

(B) Stationary phase

(C) Senescent phase

(D) Lag phase

Open solution

Q92. AIPMT 2004

In which one of the following habitats does the diurnal temperature of soil surface vary most ?

(A) Grassland

(B) Desert

(C) Shrub land

(D) Forest

Open solution

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Q93. AIPMT 2003

Two opposite forces operate in the growth and development of every population. One of them relates to the ability to reproduce at a given rate. The force opposing it is called : -

(A) Biotic potential
(B) Environmental resistance
(C) Morbidity
(D) Fecundity

Open solution

Q94. AIPMT 2003

Mycorrhiza is an example of :-

(A) Ectoparasitism
(B) Decomposers
(C) Endoparasitism
(D) Symbiotic relationship

Open solution

Q95. AIPMT 2002

The growth curve of bacterial population in lab is plotted against time. What will be the shape of graph?

(A) Sigmoid
(B) Hyperbolic
(C) Ascending straight line
(D) Descending straight line

Open solution

Q96. AIPMT 2002

In a population, unrestricted reproductive capacity is called as -

(A) Biotic potential
(B) Birth rate
(C) Carrying capacity
(D) Fertility

Open solution

Q97. AIPMT 2002

What is true for individuals of same species ?

(A) Live in same habitat
(B) Live in different habitat
(C) Interbreeding
(D) Live in same niche

Open solution

Q98. AIPMT 2002

Which type of association is found in between entomophilous flower and pollinating agent ?

(A) Cooperation
(B) Co-evolution
(C) Mutualism
(D) Commensalism

Open solution

Q99. AIPMT 2002

Two different species can not live for long duration in the same niche or habitat. This law is : -

(A) Competitive exclusion principle
(B) Weiseman's theory
(C) Allen's law
(D) Gause's hypothesis

Open solution

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Q100. AIPMT 2002

Which of the following is a correct pair?

(A) *Cuscuta* - parasite
(B) *Dischidia* - insectivorous
(C) *Opuntia* - predator
(D) *Capsella* - hydrophyte

Open solution

Q101. AIPMT 2001

What is the best pH of the soil for cultivation of plants?

(A) $3.4 - 5.4$
(B) $6.5 - 7.5$
(C) $4.5 - 8.5$
(D) $5.5 - 6.5$

Open solution

Q102. AIPMT 1999

Temperature changes in the environment affect most of the animals which are

(A) homoeothermic
(B) aquatic
(C) poikilothermic
(D) desert living

Open solution

Q103. AIPMT 1999

Angiosperm to which the largest flowers belong is

(A) partial stem parasite
(B) total stem parasite
(C) total root parasite
(D) partial root parasite

Open solution

Q104. AIPMT 1998

Two opposite forces operate in the growth and development of every population. One of them is related to the ability to reproduce at a given rate. The force opposing to it is called

(A) biotic control
(B) mortality
(C) fecundity
(D) environmental resistance

Open solution

Q105. AIPMT 1998

Plants such as *Prosopis*, *Acacia* and *Capparis* represent examples of tropical

(A) grass lands
(B) thorn forests
(C) deciduous forests
(D) evergreen forests

Open solution

Q106. AIPMT 1998

In desert grasslands, which type of animals are relatively more abundant?

(A) Diurnal
(B) Arboreal
(C) Aquatic
(D) Fossorial

Open solution

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Q107. AIPMT 1997

During adverse season, therophytes survive by

(A) rhizomes (B) seeds (C) bulbs (D) corms

Open solution

Q108. AIPMT 1996

The animals of cold countries have relatively shorter and poorly developed ears, eyes, hairs and other phenotypic characters. This is known by which law?

(A) Cope's Law (B) Dollo's Law (C) Allen's Law (D) Bergamann's Law

Open solution

Q109. AIPMT 1996

Benthoic animals are those, which

(A) are submerged in area
(B) float on the sea surface
(C) are deep dweller in sea
(D) are floating (free) organisms

Open solution

Q110. AIPMT 1996

The 'niche' of a species is meant for

(A) habitat and specific functions of a species
(B) specific place where an organism lives
(C) specific species function and its competitive power
(D) none of these

Open solution

Q111. AIPMT 1995

In India, human population is heavily weighed towards the younger age groups as a result of

(A) short life span of many individuals and high birth rate
(B) long life span of many individuals and high birth rate
(C) short life span of many individuals and low birth rate
(D) long life span of many individuals and low birth rate

Open solution

Q112. AIPMT 1995

The abundance of a species population, within its habitat, is called

(A) relative density (B) regional density (C) absolute density (D) niche density

Open solution

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Q113. AIPMT 1995

Which one of the following pairs is correctly matched?

(A) Parasitism - intraspecific relationship

(B) Uricotelism - aquatic habitat

(C) Excessive perspiration - xeric adaptation

(D) Streamlined body - aquatic adaptation

Open solution

Q114. AIPMT 1993

Study of inter-relationships between organisms and their environment is

(A) ecology (B) ecosystem (C) phytogeography (D) ethology

Open solution

Q115. AIPMT 1993

The sum total of the populations of the same kind of organisms constitute

(A) colony (B) genus (C) community (D) species

Open solution

Q116. AIPMT 1993

Association of animals when both partners are benefitted

(A) colony (B) mutualism (C) commensalism (D) ammensalism

Open solution

Q117. AIPMT 1992

Fertility of soil is measured by its ability to

(A) retain nutrients (B) hold organic materials

(C) hold water (D) support life

Open solution

Q118. AIPMT 1992

Soil particles determine its

(A) texture (B) field capacity

(C) water holding capacity (D) soil flora

Open solution

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Q119. AIPMT 1991

Homeostasis is

(A) tendency to charge with change in environment

(B) tendency to resist change

(C) disturbance in regulatory control

(D) plants and animal extracts used in homeopathy

Open solution

Q120. AIPMT 1991

Deep black soil is productive due to high proportion of

(A) sand and zinc

(B) gravel and calcium

(C) clay and humus

(D) silt and earthworm

Open solution

Q121. AIPMT 1991

Which one is true?

(A) Commensalism when none of the interacting populations affect each other.

(B) Symbiosis when the interaction is useful to both the populations.

(C) Symbiosis when neither populations affects each other.

(D) Commensalism when the interaction is useful to both the populations.

Open solution

Q122. AIPMT 1989

The relation between algae and fungi in a lichen is

(A) symbiosis

(B) parasitism

(C) commenalism

(D) protocooperation

Open solution

Q123. AIPMT 1988

Competition for light, nutrients and space is most severe between

(A) closely related organism growing in different niches

(B) closely related organisms growing in the same area/niche

(C) distantly related organisms growing in the same habitat

(D) distantly related organisms growing in different niches

Open solution

Q124. AIPMT 1988

A mutually beneficial association necessary for survival of both partners is

(A) mutualism/symbiosis

(B) commensalism

(C) amensalism

(D) both (1) and (2)

Open solution

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Answer Key

1. **B**	2. **D**	3. **B**	4. **A**	5. **B**
6. **B**	7. **C**	8. **C**	9. **D**	10. **D**
11. **D**	12. **D**	13. **B**	14. **A**	15. **C**
16. **D**	17. **D**	18. **A**	19. **B**	20. **B**
21. **A**	22. **D**	23. **D**	24. **C**	25. **C**
26. **A**	27. **A**	28. **B**	29. **B**	30. **D**
31. **D**	32. **A**	33. **C**	34. **D**	35. **D**
36. **A**	37. **D**	38. **D**	39. **D**	40. **D**
41. **B**	42. **A**	43. **B**	44. **A**	45. **D**
46. **A**	47. **C**	48. **C**	49. **A**	50. **B**
51. **B**	52. **C**	53. **D**	54. **A**	55. **D**
56. **A**	57. **A**	58. **A**	59. **D**	60. **D**
61. **D**	62. **D**	63. **A**	64. **B**	65. **C**
66. **A**	67. **B**	68. **C**	69. **B**	70. **D**
71. **C**	72. **D**	73. **A**	74. **C**	75. **D**
76. **A**	77. **D**	78. **C**	79. **A**	80. **D**
81. **D**	82. **A**	83. **A**	84. **A**	85. **A**
86. **A**	87. **C**	88. **A**	89. **B**	90. **D**
91. **A**	92. **B**	93. **B**	94. **D**	95. **A**
96. **A**	97. **C**	98. **B**	99. **D**	100. **A**
101. **B**	102. **C**	103. **C**	104. **D**	105. **B**
106. **D**	107. **B**	108. **C**	109. **C**	110. **A**
111. **A**	112. **D**	113. **D**	114. **A**	115. **D**
116. **B**	117. **D**	118. **A**	119. **B**	120. **C**
121. **B**	122. **A**	123. **B**	124. **A**	

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